

INDEPENDENT ENERGY YIELD ASSESSMENT

Hybrid EYA Report

200 MW NHPC FDRE • Tranche-II



275 MWac

Solar • 409.1 MWp



44.1 MW

Wind • 14 × 3.15 MW



400 MWh

BESS • 200 MW / 0.5C

SITE	Bikaner, Rajasthan • Bikaner ISTS GSS
CONTRACTED CAPACITY	200 MW (shared) under FDRE PPA
ASSESSMENT	P50 / P75 / P90 • 20-year horizon



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Executive Summary

Joulewise Advisory Private Limited has carried out an independent Energy Yield Assessment (EYA) of the proposed Hybrid Power Project (Solar + Wind + BESS) consisting of the following project configuration and site, for supply of Firm and Dispatchable RE power under the NHPC 1,200 MW FDRE Tranche-II tender (200 MW part capacity):

Table 1: Hybrid (Solar + Wind) Project Details

Project	Site / Location	Total Capacity	Connectivity	GSS
Solar	Bikaner, Rajasthan	275 MWac / 409.1 MWp (DC:AC 1.49)	200 MW (shared)	Bikaner ISTS GSS
Wind	Bikaner, Rajasthan	44.1 MW (14 x 3.15 MW WTG)	200 MW (shared)	Bikaner ISTS GSS
BESS	Co-located, Bikaner	400 MWh / 200 MW / C-rate 0.50	-	-
Non-Participating	Not Applicable	Nil - entire Contracted Capacity is participating under the FDRE PPA	-	-

Note:

- ◆ The entire 200 MW grid connectivity is shared across the Solar, Wind and BESS components, being the Contracted Capacity under the NHPC FDRE Tranche-II RfS.
- ◆ Only the generation meeting the Contracted Capacity is considered for FDRE compliance; energy beyond the evacuation limit is either stored in the BESS or curtailed.
- ◆ There is no Non-Participating / merchant plant component in this project configuration; the entire capacity operates under the FDRE PPA.

As per project requirement, the Consultant has conducted an independent resource assessment for both the solar and wind portions of the hybrid project and performed the hybrid analysis considering the permitted power evacuation capacity, tender conditions and PPA requirement.

Based on the overall hybrid assessment considering 200 MW PPA supply under the FDRE tender, the following is a summary of results attained from the hybrid plant including BESS:

Table 2: Hybrid EYA Results Summary (20 Years Average Values)

Sr no	Particulars	P50	P75	P90
1	Solar 275 MW Net Gen at 33 kV (GWh)	670.5	659.6	649.7
2	Wind 44.1 MW Net Gen at 33 kV (GWh)	100.5	92.4	85.1
3	Hybrid Net Gen at 33 kV (GWh)	771.0	752.0	734.8
4	RE Simultaneous Supply (GWh)	572.4	555.6	540.3
5	RE Supply to ESS (GWh)	164.2	166.4	168.3
6	RE Surplus - Power available for Merchant Sales (GWh)	0.0	0.0	0.0
7	RE Curtailment (GWh)	34.4	30.0	26.2
8	RE Transmission Loss (GWh)	14.2	13.9	13.6

Sr no	Particulars	P50	P75	P90
8a	Auxiliary Consumption (GWh)	10.7	10.4	10.2
9	ESS Supply (GWh)	138.5	140.4	142.0
10	ESS Surplus - Power available for Merchant Sales (GWh)	0.0	0.0	0.0
11	ESS RTE Loss (GWh)	25.6	26.0	26.3
12	Supply from Other Sources (GWh)	0.0	0.0	0.0
13	Energy Sold under PPA (GWh)	686.0	671.6	658.4
14	Annual CUF (%)	39.16	38.33	37.58
15	Shortfall on Annual Availability (MWh)	0.0	0.0	0.0
16	Shortfall on Monthly Availability (MWh)	2509.3	3258.9	4106.6
17	BESS 400 MWh DC Capacity (Beginning of Year 1)	400.0	400.0	400.0
18	BESS RTE (Beginning of Year 1, %)	84.6	84.6	84.6
19	Total Supply Peak (MW)	200.0	200.0	200.0
20	Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0
21	Total ESS Supply Peak (MW)	180.0	180.0	180.0
22	Total Surplus - Power available for Merchant Sales (GWh)	0.0	0.0	0.0
23	Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0
24	Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0
	January	1434.4	1652.0	1870.0
	February	268.5	407.1	560.5
	March	0.0	4.5	14.7
	April	0.0	0.0	0.0
	May	0.0	0.0	0.0
	June	0.0	0.0	0.0
	July	74.1	130.7	189.5
	August	104.6	194.6	312.5
	September	0.4	7.2	20.7
	October	0.8	4.8	16.1
	November	263.1	367.1	481.3
	December	363.5	490.8	641.4

Note: Rows shown as 0.0 (RE/ESS Surplus for Merchant Sales, Supply from Other Sources, Total Surplus variants) reflect this assessment's dispatch design, in which any generation in excess of the Contracted Capacity that cannot be stored in the BESS is treated as Curtailment rather than tracked as a separately dispatchable merchant-sale stream; the RfS permits such sale (Clause 6.3) but this has conservatively not been quantified as a separate revenue line in this EYA.



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Introduction

Joulewise Advisory Private Limited has been engaged to carry out an independent Energy Yield Assessment (EYA) of the proposed Hybrid Power Project under FDRE Tender for 200 MW PPA capacity, consisting of Solar and Wind power projects including BESS. Project configuration is given in the table below:

Table 3: Hybrid (Solar + Wind) Project Details

Project	Site / Location	Total Capacity	Connectivity	GSS
Solar	Bikaner, Rajasthan	275 MWac / 409.1 MWp (DC:AC 1.49)	200 MW (shared)	Bikaner ISTS GSS
Wind	Bikaner, Rajasthan	44.1 MW (14 x 3.15 MW WTG)	200 MW (shared)	Bikaner ISTS GSS
BESS	Co-located, Bikaner	400 MWh / 200 MW / C-rate 0.50	-	-
Non-Participating	Not Applicable	Nil - entire Contracted Capacity is participating under the FDRE PPA	-	-

As per project requirement, the Consultant has conducted an independent resource assessment for both the solar and wind portions of the hybrid project and performed the hybrid analysis considering permitted power evacuation capacity, tender conditions and PPA requirement.

A brief EYA summary of the individual Wind and Solar project sites along with the Hybrid project EYA is given in the following sections of the report.



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Wind Eya Summary

Joulewise Advisory Private Limited has carried out an independent Energy Yield Assessment (EYA) of the proposed wind farm located in the Bikaner region of Rajasthan.

The windfarm is comprised of a total of 14 Wind Turbine Generators (WTGs) of SUZLON S144 3.15MW AT 140m hub height. Coordinates of individual proposed WTG positions are given in the below table:

Table 4: WTG Coordinates

Turbine ID	Easting (m)	Northing (m)	Zone (UTM)	Mast Association	Elevation (m)
BKNRJ501	335000	3101000	43R	Bikaner	228
BKNRJ502	335480	3100940	43R	Bikaner	231
BKNRJ503	335960	3100880	43R	Bikaner	235
BKNRJ504	336440	3100820	43R	Bikaner	238
BKNRJ505	336920	3100760	43R	Bikaner	241
BKNRJ506	337400	3100700	43R	Bikaner	244
BKNRJ507	337880	3100640	43R	Bikaner	240
BKNRJ508	335220	3101950	43R	Bikaner	236
BKNRJ509	335700	3101890	43R	Bikaner	233
BKNRJ510	336180	3101830	43R	Bikaner	229

Turbine ID	Easting (m)	Northing (m)	Zone (UTM)	Mast Association	Elevation (m)
BKNRJ511	336660	3101770	43R	Bikaner	232
BKNRJ512	337140	3101710	43R	Bikaner	236
BKNRJ513	337620	3101650	43R	Bikaner	239
BKNRJ514	338100	3101590	43R	Bikaner	242

Note: The above WTG coordinates are indicative/proposed, generated on the basis of a typical turbine spacing layout for the Bikaner site; exact positions are subject to finalization by the developer through micro-siting survey and wind resource mast data, which were not available as source data for this assessment.

A summary of project and Annual Energy Production (AEP) results is given in below tables:

Table 5: Summary of Wind Resource Assessment

1	WTG Model	-	Suzlon S144_3.15MW (assumed basis)
2	Rotor Diameter (m)	-	144
3	Hub Height (m)	-	140
4	WTG Capacity (MW)	-	3.15
5	No. of WTGs	-	14
6	Project Capacity (MW)	-	44.10
7	Total Loss without wake	-	10.25%
8	Total Loss with wake	-	24.35%
9	Uncertainty (25 years)	-	9.47%
10	Net Generation (MWh/MW/year)	P50	2,322.9
		P75	2,134.9
		P90	1,965.6
11	Net PLF (%)	P50	26.52%
		P75	24.37%
		P90	22.44%

Table 6: Uncertainty Values (S144 3.15 MW at 140 m hub height)

Type of Uncertainty	Uncertainty on AEP (25 Years) (%)
Wind Measurement uncertainty	2.51%
Data Documentation uncertainty	1.00%
Long Term scaling uncertainty	2.00%
Wind resource variability	1.41%
Vertical extrapolation uncertainty	4.00%
Horizontal extrapolation uncertainty	3.75%
Wake modelling	5.50%
Power Curve uncertainty	4.00%

Type of Uncertainty	Uncertainty on AEP (25 Years) (%)
Total	9.47%

Note: The wind resource dataset used represents the Bikaner site on a per-installed-MW basis (loss cascade consistent with a Suzlon S144 3.15 MW class turbine) and has been scaled to the selected 44.1 MW capacity (14 turbines).

4 Solar Eya Summary

The Hybrid Configuration includes a Solar Power component of 275 MWac (409.1 MWp) capacity at the Bikaner site. Brief details of the Solar EYA are given below:

4.1. Site – Bikaner (275 MWac)

The Solar PV Project is proposed to be developed at Bikaner, Rajasthan, India. Independent Energy Yield Assessment (EYA) of the Project was carried out using PVsyst V8.1.4 software, referencing a 360 MWp / 242 MWac PVsyst simulation run for the Bikaner site (Bifacial modules, fixed tilt, 29 deg tilt angle), scaled linearly to the selected 275 MWac capacity.

Table 7: Summary of Project & Annual Energy Production (275 MWac)

Project Capacity	409.1 MWp / 275 MWac
DC:AC Ratio	1.49
Location	Bikaner, Rajasthan
Latitude	28.3800 N
Longitude	73.1600 E
Altitude	191 m
Meteo Dataset Source	PVsyst Meteonorm 9.0 (Sat=100%, Synthetic)
Global Horizontal Irradiance (GHI)	1,984.4 kWh/m ² /year
Annual Average Ambient Temp	27.23 deg C
SPV Technology	Bifacial (Generic BiN-02-610, 610 Wp)
Approx. No. of Modules (scaled)	670,636
No. of Modules per String	30
Total No. of Strings (scaled)	22,355
Inverter Manufacturer / Rating	Generic SG4400-UD-MV-US, 4,400 kWac
Approx. No. of Inverters (scaled)	63
MMS Type & Tilt Angle	Fixed Tilt, 29 deg
Pitch & Collector Width	6.00 m & 4.40 m
P50 Generation (MWh) - 1st Year, Solar only	696,369
P75 Generation (MWh) - 1st Year, Solar only	685,006
P90 Generation (MWh) - 1st Year, Solar only	674,756
DC CUF (%) - P50	19.43%
AC CUF (%) - P50	28.91%

Project Capacity	409.1 MWp / 275 MWac
Performance Ratio (PR) - P50 (PVsyst source run)	76.05%

Note: Equipment counts (module/inverter/string quantities) are scaled pro-rata from the PVsyst source run and are indicative; exact module/inverter selection, string sizing and layout are subject to detailed engineering by the developer.

5 Battery Energy Storage System

A Battery Energy Storage System (BESS) has been considered for hybrid project supply. The following are the technical details/inputs of the BESS used for the Hybrid Assessment:

Table 8: 400 MWh BESS Technical Inputs

Sr.	Input Parameters	Unit	Value
1	BESS Nameplate Capacity	MWh/MW	400 MWh / 200 MW
2	No. of Containers	Nos	80
3	Each Container Capacity	MWh	5.0
4	Storage Duration (Nameplate)	hours	2.0
5	Charge Efficiency	%	92.0%
6	Discharge Efficiency	%	92.0%
7	Round Trip Efficiency (RTE), Year 1	%	84.64%
8	Cycle per Day	Nos	1
9	DoD of usable capacity	%	100%
10	Battery Augmentation	Year	100% SoH restored, 11th year
11	Usable Capacity Formula	-	Nameplate (MWh) x SoH (%) x DoD (%) – applied year-specifically in dispatch simulation
12	Auxiliary Consumption	% of Gross Supply/ year	1.5%

Table 9: 400 MWh BESS SoH and RTE Schedule

Yr	DC Usable (MWh)	SoH (%)	RTE (%)
1	400.0	100.00%	84.64%
2	384.3	96.08%	84.51%
3	376.3	94.07%	84.46%
4	369.6	92.40%	84.42%
5	363.6	90.91%	84.39%
6	358.2	89.54%	84.36%
7	353.1	88.27%	84.34%
8	348.3	87.07%	84.31%

Yr	DC Usable (MWh)	SoH (%)	RTE (%)
9	343.7	85.93%	84.29%
10	339.4	84.84%	84.27%
11	400.0	100.00%	84.64%
12	384.3	96.08%	84.51%
13	376.3	94.07%	84.46%
14	369.6	92.40%	84.42%
15	363.6	90.91%	84.39%
16	358.2	89.54%	84.36%
17	353.1	88.27%	84.34%
18	348.3	87.07%	84.31%
19	343.7	85.93%	84.29%
20	339.4	84.84%	84.27%

¹ DoD at the system level is provided by the OEM as 100%. This is a commercial construct.



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Hybrid Eya Summary

6.1. Methodology

The study is carried out to understand the revenue potential of the proposed project. A rule-based, day-ahead-aware dispatch simulation is used to model the hybrid plant's 15-minute generation, BESS charge/discharge behaviour and settlement against the tender's Firm and Dispatchable RE power supply conditions. Main components of the assessment are as follows:

A. OBJECTIVE – MAXIMIZE REVENUE GENERATING ENERGY

Revenue generating energy = Energy Supplied under PPA – 1.5 × Energy Shortfall under PPA + 1.0 × Energy Surplus

Note:

- ◆ Since each unit of energy shortfall is penalized at 1.5 times the tariff, the weightage considered for energy shortfall is as per the PPA terms.
- ◆ In order to prioritize PPA supply over the surplus, the surplus is given a lower weightage.

B. CONSTRAINTS

- ◆ Annual basis: (a) Minimum CUF under the PPA – any supply below the minimum CUF will be penalized as per PPA terms; (b) Maximum CUF under the PPA – any supply above the maximum CUF will not be purchased under the PPA.
- ◆ Monthly basis: Minimum monthly availability during the peak hours – any supply below the minimum monthly availability during peak hours will be penalized as per PPA terms.
- ◆ Hourly/15-min basis: (a) Supply under PPA cannot be more than the PPA capacity in any time block; (b) Total evacuation, i.e. sum of supply and surplus, cannot be higher than the evacuation capacity.

C. INPUTS

- ◆ First-year 15-minute generation profiles of Solar at P50, P75 and P90 (synthesized from the PVsyst monthly output, scaled to 275 MWac). Using a deration factor of 0.4%/year, hourly/15-min generation profiles are calculated for the rest of the project life.

- ◆ First-year 15-minute generation profiles of Wind at P50, P75 and P90 (interpolated from the source hourly resource data, scaled to 44.1 MW), degraded year-on-year at 0.20%/year over the 25-year PPA term.
- ◆ Profile of BESS including initial capacity, charging and discharging rates, losses in percentage, depth of discharge, State-of-Health schedule for life of the battery, round trip energy schedule. BESS Usable Capacity (MWh) for each year = Nameplate Capacity (400 MWh) x State-of-Health (%) x Depth-of-Discharge (%); this year-specific usable capacity (not the flat nameplate value) is used as the operative energy limit in the dispatch simulation for that year, together with the corresponding year-specific Round-Trip Efficiency.
- ◆ A flat Auxiliary Consumption of 1.5% per year is deducted from Gross Supply (Simultaneous Supply + ESS Supply) to arrive at Net Supply under the PPA, representing the hybrid plant's own station power/parasitic load.

D. ASSESSMENT

Assessment is carried out at P50, P75 and P90. Energy generated by the solar and wind is distributed among the following components:

- ◆ Simultaneous Supply – The energy supplied immediately to the buyer under the PPA is called Simultaneous Supply.
- ◆ Supply to Energy Storage Solution (ESS) – A part of the energy generated is supplied to ESS to meet the demand when the generation is unable to meet the peak period demand. A part of this energy is lost due to internal transmission losses and energy conversion losses of the ESS. The rest is used for charging the ESS. The energy that ultimately charges the battery (i.e., after the losses) is termed as Energy for ESS Charging.
- ◆ Surplus – The generated energy that is in excess to (i) Simultaneous Supply and Supply to ESS on an hourly basis within contracted capacity, and (ii) upper limit of supply allowed under PPA, is the surplus which is available for sale to other buyers.
- ◆ Curtailment – The energy that cannot be evacuated due to the evacuation limit or stored in the ESS is called Curtailment.
- ◆ Losses – Losses are the ESS Transmission Loss and ESS Conversion Loss. ESS Transmission Loss is the loss of energy due to transmission of AC power (i) from the generation busbar to the ESS and (ii) from the ESS to the Point of Injection. ESS Conversion Loss is the loss of energy due to (i) conversion AC to DC and (ii) conversion DC to AC and (iii) losses within the ESS.

The hybrid generation is the sum of the above five components.

Supply

Energy supply under the PPA has two components:

- ◆ Simultaneous Supply – Explained under Generation and Utilization section above.
- ◆ ESS Supply – Energy supplied by the ESS to meet the supply obligations under the PPA.

6.2. Supply Conditions and Revenue Components

The revenue generated from the proposed project is from three components:

- ◆ Supply under the Power Purchase Agreement (PPA). As per PPA, the proposed project should meet the following conditions: (a) Annual CUF of the proposed project should be between -15% and +10% of the declared CUF; (b) Monthly availability of the proposed plant should be 90% during the peak hours; (c) Tariff is common for both peak as well as non-peak hours.
- ◆ Sale of surplus to other buyers. Surplus can be: (a) Unused generation over and above the PPA capacity on the hourly basis; (b) Unused generation over and above the annual CUF + 10% on annual basis.
- ◆ Penalty for shortfall on the supply as per the PPA supply conditions. This component has a negative impact on the revenue. Penalty is calculated as 1.5 times the tariff for the shortfall energy.

Peak hours: 2 hours within 05:00-10:00 and 2 hours within 18:00-23:00 daily (4 hours total), as scheduled by the Buying Entity. This assessment adopts an illustrative schedule of 08:00-10:00 and 18:00-20:00 pending the actual notified schedule.

6.3. Hybrid Project Results

Considering the FDRE Tender requirement, permitted power evacuation capacity, BESS utilization, clipping analysis, generation profile and other technical details, a detailed Hybrid Generation analysis has been performed on different probability values of generation of both Wind and Solar energy.

For Hybridization, year-on-year degraded hourly generation profiles were used for both the Wind (0.20%/year) and Solar (0.40%/year) power plants over the 20-year assessment period, together with a year-specific BESS usable capacity (Nameplate x SoH x DoD) and Round-Trip Efficiency, and a flat 1.5%/year Auxiliary Consumption deduction.

Following table provides the 20-year average generation values used for assessment:

Table 13: Hybrid EYA Results Summary (20 Years Average Values)

Sr no	Particulars	P50	P75	P90
1	Solar 275 MW Net Gen at 33 kV (GWh)	670.5	659.6	649.7
2	Wind 44.1 MW Net Gen at 33 kV (GWh)	100.5	92.4	85.1
3	Hybrid Net Gen at 33 kV (GWh)	771.0	752.0	734.8
4	RE Simultaneous Supply (GWh)	572.4	555.6	540.3
5	RE Supply to ESS (GWh)	164.2	166.4	168.3
6	RE Surplus - Power available for Merchant Sales (GWh)	0.0	0.0	0.0
7	RE Curtailment (GWh)	34.4	30.0	26.2
8	RE Transmission Loss (GWh)	14.2	13.9	13.6
8a	Auxiliary Consumption (GWh)	10.7	10.4	10.2
9	ESS Supply (GWh)	138.5	140.4	142.0
10	ESS Surplus - Power available for Merchant Sales (GWh)	0.0	0.0	0.0
11	ESS RTE Loss (GWh)	25.6	26.0	26.3
12	Supply from Other Sources (GWh)	0.0	0.0	0.0
13	Energy Sold under PPA (GWh)	686.0	671.6	658.4
14	Annual CUF (%)	39.16	38.33	37.58
15	Shortfall on Annual Availability (MWh)	0.0	0.0	0.0
16	Shortfall on Monthly Availability (MWh)	2509.3	3258.9	4106.6
17	BESS 400 MWh DC Capacity (Beginning of Year 1)	400.0	400.0	400.0
18	BESS RTE (Beginning of Year 1, %)	84.6	84.6	84.6
19	Total Supply Peak (MW)	200.0	200.0	200.0
20	Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0
21	Total ESS Supply Peak (MW)	180.0	180.0	180.0
22	Total Surplus - Power available for Merchant Sales (GWh)	0.0	0.0	0.0
23	Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0
24	Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0
	January	1434.4	1652.0	1870.0
	February	268.5	407.1	560.5

Sr no	Particulars	P50	P75	P90
	March	0.0	4.5	14.7
	April	0.0	0.0	0.0
	May	0.0	0.0	0.0
	June	0.0	0.0	0.0
	July	74.1	130.7	189.5
	August	104.6	194.6	312.5
	September	0.4	7.2	20.7
	October	0.8	4.8	16.1
	November	263.1	367.1	481.3
	December	363.5	490.8	641.4

The Hybrid EYA Results Summary above (Table 13) reproduces Table 2 of the Executive Summary, presented here in accordance with the standard EYA report structure.

Considering the FDRE Tender requirement, permitted power evacuation capacity, BESS utilization and other technical details, a detailed hybrid generation analysis has been performed on different probability values of generation of both Solar and Wind energy. The following are the monthly results for each probability case (Year 1):

Table 14: P50 Monthly Energy Settlement (Year 1)

Month	RE Gen (MWh)	Simult. Supply (MWh)	ESS Charge-In (MWh)	ESS RTE Loss (MWh)	ESS Supply (MWh)	Curtail- ment (MWh)	Tx Loss (MWh)	Aux Consum. (MWh)	Gross Supply (MWh)	Net PPA Supply (MWh)	Req. Peak Energy (MWh)	Actual Peak Supply (MWh)	Peak Avail (%)
January	60506	41603	15902	2443	13367	3001	1099	825	54970	53046	22320	21769	97.5
February	61615	43361	13714	2106	11641	4540	1100	825	55002	53077	20160	20292	100.7
March	72631	52793	13932	2140	11796	5906	1292	969	64589	62328	22320	22719	101.8
April	73849	56057	11996	1843	10123	5797	1324	993	66180	63864	21600	22170	102.6
May	73959	59530	11844	1819	9989	2586	1390	1043	69519	67086	22320	22713	101.8
June	73281	60609	10927	1678	9230	1746	1397	1048	69839	67395	21600	21946	101.6
July	68846	56383	12463	1914	10660	0	1341	1006	67043	64696	22320	22367	100.2
August	62752	48657	14095	2165	11928	0	1212	909	60585	58464	22320	22321	100.0
September	64680	48296	13444	2065	11351	2940	1193	895	59646	57559	21600	21846	101.1
October	65661	47655	14215	2183	12048	3791	1194	896	59703	57614	22320	22722	101.8
November	59392	42789	14614	2245	12318	1989	1102	827	55106	53177	21600	21715	100.5
December	61634	44638	15379	2362	13014	1617	1153	865	57653	55635	22320	22308	99.9
ANNUAL TOTAL	798807	602370	162525	24964	137465	33912	14797	11098	739835	713941	262800	264887	

Table 15: P75 Monthly Energy Settlement (Year 1)

Month	RE Gen (MWh)	Simult. Supply (MWh)	ESS Charge-In (MWh)	ESS RTE Loss (MWh)	ESS Supply (MWh)	Curtail- ment (MWh)	Tx Loss (MWh)	Aux Consum. (MWh)	Gross Supply (MWh)	Net PPA Supply (MWh)	Req. Peak Energy (MWh)	Actual Peak Supply (MWh)	Peak Avail (%)
January	59072	40476	16000	2458	13454	2596	1079	809	53930	52043	22320	21653	97.0
February	60073	42081	13932	2140	11823	4060	1078	809	53904	52017	20160	20254	100.5
March	70950	51543	14139	2172	11971	5268	1270	953	63513	61290	22320	22690	101.7
April	72105	54844	12201	1874	10299	5060	1303	977	65143	62863	21600	22128	102.4
May	72142	58047	12115	1861	10221	1981	1365	1024	68268	65878	22320	22661	101.5
June	71362	58941	11224	1724	9483	1198	1369	1026	68423	66029	21600	21891	101.3
July	66753	53922	12831	1971	10974	0	1298	973	64896	62625	22320	22321	100.0
August	61156	46763	14393	2211	12178	0	1179	884	58942	56879	22320	22303	99.9
September	63303	47208	13656	2098	11524	2439	1175	881	58732	56676	21600	21810	101.0
October	64368	46674	14376	2208	12184	3318	1177	883	58858	56798	22320	22695	101.7
November	57947	41538	14825	2277	12500	1585	1081	811	54038	52146	21600	21675	100.3
December	59925	43040	15646	2403	13241	1239	1126	844	56281	54311	22320	22273	99.8
ANNUAL TOTAL	779157	585076	165339	25396	139851	28742	14499	10874	724927	699554	262800	264354	

Table 16: P90 Monthly Energy Settlement (Year 1)

Month	RE Gen (MWh)	Simult. Supply (MWh)	ESS Charge-In (MWh)	ESS RTE Loss (MWh)	ESS Supply (MWh)	Curtail- ment (MWh)	Tx Loss (MWh)	Aux Consum. (MWh)	Gross Supply (MWh)	Net PPA Supply (MWh)	Req. Peak Energy (MWh)	Actual Peak Supply (MWh)	Peak Avail (%)
January	57779	39453	16088	2471	13532	2238	1060	795	52985	51131	22320	21547	96.5
February	58683	40914	14123	2169	11982	3645	1058	793	52897	51045	20160	20210	100.2
March	69434	50417	14327	2201	12129	4690	1251	938	62545	60356	22320	22654	101.5
April	70533	53692	12394	1904	10463	4447	1283	962	64156	61910	21600	22089	102.3
May	70504	56665	12369	1900	10438	1470	1342	1007	67102	64754	22320	22613	101.3
June	69632	57386	11501	1767	9718	744	1342	1007	67105	64756	21600	21840	101.1
July	64866	51729	13137	2018	11235	0	1259	945	62964	60761	22320	22258	99.7
August	59717	45096	14621	2246	12371	0	1149	862	57467	55455	22320	22254	99.7
September	62062	46206	13850	2127	11682	2006	1158	868	57888	55862	21600	21769	100.8
October	63202	45762	14523	2231	12307	2916	1161	871	58070	56037	22320	22664	101.5
November	56645	40376	15018	2307	12668	1251	1061	796	53043	51187	21600	21642	100.2
December	58384	41598	15879	2439	13438	908	1101	826	55035	53109	22320	22235	99.6
ANNUAL TOTAL	761440	569293	167830	25779	141964	24316	14225	10669	711257	686363	262800	263774	

 Annexure 1: Yearly Results

P50 – Yearly Results (20 Years)

Years 1-10

Particulars	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
Solar 275 MW Net Gen at 33 kV (GWh)	696.4	693.6	690.8	688.0	685.3	682.6	679.8	677.1	674.4	671.7
Wind 44.1 MW Net Gen at 33 kV (GWh)	102.4	102.2	102.0	101.8	101.6	101.4	101.2	101.0	100.8	100.6
Hybrid Net Gen at 33 kV (GWh)	798.8	795.8	792.8	789.9	786.9	784.0	781.0	778.1	775.2	772.3
RE Simultaneous Supply (GWh)	602.4	595.8	591.3	587.4	583.8	580.4	577.1	573.8	570.8	568.2
RE Supply to ESS (GWh)	162.5	162.7	162.9	162.9	162.9	162.9	162.8	162.7	162.5	162.2
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	33.9	37.3	38.6	39.5	40.2	40.7	41.2	41.6	41.9	41.9
RE Transmission Loss (GWh)	14.8	14.7	14.6	14.5	14.4	14.4	14.3	14.2	14.2	14.1
Auxiliary Consumption (GWh)	11.1	11.0	10.9	10.9	10.8	10.8	10.7	10.7	10.6	10.6
ESS Supply (GWh)	137.5	137.4	137.5	137.5	137.4	137.4	137.3	137.1	136.9	136.7
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	25.0	25.2	25.3	25.4	25.4	25.5	25.5	25.5	25.5	25.5
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	713.9	707.6	703.3	699.5	696.0	692.6	689.3	686.0	683.0	680.2
Annual CUF (%)	40.75	40.39	40.14	39.93	39.72	39.53	39.35	39.16	38.98	38.82
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	563.1	866.3	1070.5	1332.4	1697.6	2078.6	2488.2	2917.8	3354.9	3817.2
BESS 400 MWh DC Capacity (Beg. of Year, MWh)	400.0	384.3	376.3	369.6	363.6	358.2	353.1	348.3	343.7	339.4
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Years 11-20

Particulars	Yr 11	Yr 12	Yr 13	Yr 14	Yr 15	Yr 16	Yr 17	Yr 18	Yr 19	Yr 20
Solar 275 MW Net Gen at 33 kV (GWh)	669.0	666.3	663.7	661.0	658.4	655.7	653.1	650.5	647.9	645.3
Wind 44.1 MW Net Gen at 33 kV (GWh)	100.4	100.2	100.0	99.8	99.6	99.4	99.2	99.0	98.8	98.6
Hybrid Net Gen at 33 kV (GWh)	769.4	766.5	763.7	760.8	758.0	755.1	752.3	749.5	746.7	743.9
RE Simultaneous Supply (GWh)	580.6	574.0	569.7	565.8	562.3	559.0	555.8	552.8	549.9	547.3
RE Supply to ESS (GWh)	166.1	166.1	166.1	166.0	165.9	165.8	165.6	165.4	165.1	164.8
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	22.8	26.4	27.9	29.0	29.8	30.4	30.9	31.3	31.7	31.9
RE Transmission Loss (GWh)	14.4	14.3	14.2	14.1	14.0	14.0	13.9	13.8	13.8	13.7
Auxiliary Consumption (GWh)	10.8	10.7	10.6	10.6	10.5	10.5	10.4	10.4	10.3	10.3
ESS Supply (GWh)	140.5	140.3	140.2	140.1	139.9	139.8	139.6	139.4	139.1	138.8
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	25.5	25.7	25.8	25.9	25.9	25.9	25.9	25.9	25.9	25.9
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	695.8	689.3	685.1	681.2	677.7	674.3	671.1	667.9	664.9	662.1
Annual CUF (%)	39.71	39.34	39.10	38.88	38.68	38.49	38.31	38.12	37.95	37.79
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	942.6	1386.6	1830.0	2278.8	2732.4	3198.8	3668.6	4158.2	4646.0	5158.0
BESS 400 MWh DC Capacity (Beg. of Year, MWh)	400.0	384.3	376.3	369.6	363.6	358.2	353.1	348.3	343.7	339.4
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Monthly Shortage (20-year average)

Monthly Shortage (MWh)	January	February	March	April	May	June	July	August	September	October	November	December
20-yr Average	1434.4	268.5	0.0	0.0	0.0	0.0	74.1	104.6	0.4	0.8	263.1	363.5

P75 – Yearly Results (20 Years)

Years 1-10

Particulars	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
Solar 275 MW Net Gen at 33 kV (GWh)	685.0	682.3	679.5	676.8	674.1	671.4	668.7	666.1	663.4	660.7
Wind 44.1 MW Net Gen at 33 kV (GWh)	94.2	94.0	93.8	93.6	93.4	93.2	93.0	92.8	92.7	92.5
Hybrid Net Gen at 33 kV (GWh)	779.2	776.2	773.3	770.4	767.5	764.6	761.8	758.9	756.0	753.2
RE Simultaneous Supply (GWh)	585.1	578.5	574.2	570.3	566.8	563.5	560.3	557.2	554.4	551.8
RE Supply to ESS (GWh)	165.3	165.4	165.4	165.4	165.3	165.2	165.0	164.7	164.4	164.1
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	28.7	32.3	33.7	34.7	35.4	36.0	36.4	36.9	37.3	37.3
RE Transmission Loss (GWh)	14.5	14.4	14.3	14.2	14.1	14.1	14.0	13.9	13.9	13.8
Auxiliary Consumption (GWh)	10.9	10.8	10.7	10.6	10.6	10.5	10.5	10.4	10.4	10.4
ESS Supply (GWh)	139.9	139.7	139.6	139.5	139.4	139.3	139.1	138.9	138.6	138.3
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	25.4	25.6	25.7	25.8	25.8	25.8	25.8	25.8	25.8	25.8
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	699.6	693.1	688.8	685.0	681.5	678.2	675.0	671.7	668.7	665.9
Annual CUF (%)	39.93	39.56	39.32	39.10	38.90	38.71	38.53	38.34	38.17	38.01
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	730.7	1095.3	1475.9	1917.3	2368.3	2837.3	3313.6	3814.4	4309.5	4809.8
BESS 400 MWh DC Capacity (Beg. of Year, MWh)	400.0	384.3	376.3	369.6	363.6	358.2	353.1	348.3	343.7	339.4
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Years 11-20

Particulars	Yr 11	Yr 12	Yr 13	Yr 14	Yr 15	Yr 16	Yr 17	Yr 18	Yr 19	Yr 20
Solar 275 MW Net Gen at 33 kV (GWh)	658.1	655.5	652.8	650.2	647.6	645.0	642.5	639.9	637.3	634.8
Wind 44.1 MW Net Gen at 33 kV (GWh)	92.3	92.1	91.9	91.7	91.5	91.4	91.2	91.0	90.8	90.6
Hybrid Net Gen at 33 kV (GWh)	750.4	747.6	744.8	742.0	739.2	736.4	733.6	730.9	728.1	725.4
RE Simultaneous Supply (GWh)	563.3	557.0	552.7	548.9	545.4	542.1	539.1	536.2	533.5	531.0
RE Supply to ESS (GWh)	168.7	168.5	168.4	168.3	168.1	167.9	167.6	167.3	166.9	166.5
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	18.4	22.0	23.6	24.8	25.7	26.4	26.9	27.3	27.8	27.9
RE Transmission Loss (GWh)	14.1	14.0	13.9	13.8	13.7	13.7	13.6	13.5	13.5	13.4
Auxiliary Consumption (GWh)	10.6	10.5	10.4	10.4	10.3	10.3	10.2	10.2	10.1	10.1
ESS Supply (GWh)	142.7	142.4	142.2	142.0	141.8	141.6	141.3	141.0	140.7	140.3
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	25.9	26.1	26.2	26.2	26.2	26.3	26.2	26.2	26.2	26.2
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	681.3	674.9	670.6	666.7	663.1	659.8	656.7	653.5	650.5	647.8
Annual CUF (%)	38.89	38.52	38.27	38.05	37.85	37.66	37.48	37.30	37.13	36.97
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	1280.2	1977.3	2517.5	3037.1	3555.6	4091.9	4625.6	5162.3	5745.8	6512.3
BESS 400 MWh DC Capacity (Beg. of Year, MWh)	400.0	384.3	376.3	369.6	363.6	358.2	353.1	348.3	343.7	339.4
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Monthly Shortage (20-year average)

Monthly Shortage (MWh)	January	February	March	April	May	June	July	August	September	October	November	December
20-yr Average	1652.0	407.1	4.5	0.0	0.0	0.0	130.7	194.6	7.2	4.8	367.1	490.8

P90 – Yearly Results (20 Years)

Years 1-10

Particulars	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
Solar 275 MW Net Gen at 33 kV (GWh)	674.8	672.1	669.4	666.7	664.0	661.4	658.7	656.1	653.5	650.8
Wind 44.1 MW Net Gen at 33 kV (GWh)	86.7	86.5	86.3	86.2	86.0	85.8	85.6	85.5	85.3	85.1
Hybrid Net Gen at 33 kV (GWh)	761.4	758.6	755.7	752.9	750.0	747.2	744.4	741.6	738.8	736.0
RE Simultaneous Supply (GWh)	569.3	562.9	558.6	554.8	551.4	548.1	545.1	542.1	539.3	536.9
RE Supply to ESS (GWh)	167.8	167.7	167.6	167.4	167.2	167.0	166.8	166.5	166.1	165.7
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	24.3	28.0	29.5	30.6	31.4	32.0	32.5	32.9	33.3	33.4
RE Transmission Loss (GWh)	14.2	14.1	14.0	13.9	13.8	13.8	13.7	13.6	13.6	13.5
Auxiliary Consumption (GWh)	10.7	10.6	10.5	10.4	10.4	10.3	10.3	10.2	10.2	10.1
ESS Supply (GWh)	142.0	141.6	141.5	141.3	141.1	140.9	140.6	140.3	140.0	139.6
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	25.8	26.0	26.0	26.1	26.1	26.1	26.1	26.1	26.1	26.1
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	686.4	679.9	675.6	671.7	668.2	664.9	661.7	658.6	655.6	652.9
Annual CUF (%)	39.18	38.81	38.56	38.34	38.14	37.95	37.77	37.59	37.42	37.26
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	986.6	1561.2	2095.7	2610.1	3132.0	3667.4	4202.1	4740.9	5276.4	5864.3
BESS 400 MWh DC Capacity (Beg. of Year, MWh)	400.0	384.3	376.3	369.6	363.6	358.2	353.1	348.3	343.7	339.4
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Years 11-20

Particulars	Yr 11	Yr 12	Yr 13	Yr 14	Yr 15	Yr 16	Yr 17	Yr 18	Yr 19	Yr 20
Solar 275 MW Net Gen at 33 kV (GWh)	648.2	645.7	643.1	640.5	637.9	635.4	632.8	630.3	627.8	625.3
Wind 44.1 MW Net Gen at 33 kV (GWh)	85.0	84.8	84.6	84.5	84.3	84.1	84.0	83.8	83.6	83.4
Hybrid Net Gen at 33 kV (GWh)	733.2	730.4	727.7	725.0	722.2	719.5	716.8	714.1	711.4	708.7
RE Simultaneous Supply (GWh)	547.4	541.4	537.3	533.5	530.1	527.0	524.1	521.3	518.7	516.4
RE Supply to ESS (GWh)	171.1	170.7	170.4	170.2	169.9	169.6	169.3	168.8	168.3	167.8
RE Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RE Curtailment (GWh)	14.8	18.4	20.0	21.3	22.2	22.9	23.4	23.9	24.4	24.5
RE Transmission Loss (GWh)	13.8	13.7	13.6	13.5	13.5	13.4	13.3	13.3	13.2	13.2
Auxiliary Consumption (GWh)	10.4	10.3	10.2	10.2	10.1	10.1	10.0	10.0	9.9	9.9
ESS Supply (GWh)	144.7	144.2	143.9	143.6	143.3	143.0	142.7	142.3	141.9	141.4
ESS Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
ESS RTE Loss (GWh)	26.3	26.4	26.5	26.5	26.5	26.5	26.5	26.5	26.4	26.4
Supply from Other Sources (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Energy Sold under PPA (GWh)	667.8	661.6	657.3	653.4	649.9	646.6	643.5	640.4	637.5	634.8
Annual CUF (%)	38.12	37.76	37.52	37.29	37.09	36.90	36.73	36.55	36.38	36.23
Shortfall on Annual Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shortfall on Monthly Availability (MWh)	1752.5	2634.4	3255.8	3857.5	4453.7	5043.5	5633.4	6283.8	7095.5	7984.8
BESS 400 MWh DC Capacity (Beg. of Year, MWh)	400.0	384.3	376.3	369.6	363.6	358.2	353.1	348.3	343.7	339.4
BESS RTE (Beg. of Year, %)	84.6	84.5	84.5	84.4	84.4	84.4	84.3	84.3	84.3	84.3
Total Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total Simultaneous Supply Peak (MW)	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0	200.0
Total ESS Supply Peak (MW)	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0	180.0
Total Surplus - Merchant Sales (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen Over Contract Cap) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total Surplus (Gen within Contract Capacity) (GWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Monthly Shortage (20-year average)

Monthly Shortage (MWh)	January	February	March	April	May	June	July	August	September	October	November	December
20-yr Average	1870.0	560.5	14.7	0.0	0.0	0.0	189.5	312.5	20.7	16.1	481.3	641.4

Note: Wind energy generation was initially estimated at the 220 kV grid interconnection level, which inherently includes losses associated with transmission lines and power transformers. The generation was subsequently adjusted to the 33 kV level by excluding the transmission and transformer losses. In contrast, the solar generation was directly estimated at the 33 kV level. The solar energy available at the 33 kV level was then considered as the input for battery charging, and a rule-based dispatch approach was employed to determine the optimal battery charging and estimate the overall hybrid energy generation.

Note on methodology: (i) Solar 15-minute generation shape is synthetic (sunrise/sunset-based, inverter-capped), scaled to reproduce the PVsyst-reported monthly energy exactly; P75/P90 monthly shapes are derived by applying the uniform annual P75/P90-to-P50 ratio, as no monthly P75/P90 split was available from the source PVsyst report. (ii) Wind per-MW basis assumes 1 WTG = 3.15 MW; per-MW yield is scaled to the selected 44.1 MW capacity. (iii) Solar degradation of 0.4%/year and Wind degradation of 0.20%/year are applied year-on-year over the 20-year assessment. (iv) BESS: 400 MWh is the Nameplate Capacity; the Usable Capacity applied in the dispatch simulation for each year = Nameplate x SoH (%) x DoD (%), together with the corresponding year-specific Round-Trip Efficiency (84.64% in Year 1, degrading per the SoH schedule); SoH schedule assumes battery augmentation restoring 100% SoH in Year 11. (v) A flat Auxiliary Consumption of 1.5%/year is deducted from Gross Supply to arrive at Net Supply under the PPA. (vi) Monthly Peak Availability is assessed on an ENERGY-RATIO basis (Actual Peak-Hour Energy Supplied / Required Peak-Hour Energy per month) consistent with the RfS provision that shortfall is calculated at the end of each month.